

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
Student Name:	K.Vijay Bhaskar Reddy	Reg. No.:	192425155
Section / Batch:	AIML/2024	Date:	28/08/2026

Code of Conduct

I K.VijayBhaskar Reddy(192425155) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Depth Estimation Comparison – Stereo Vision vs Structure from Motion

Introduction

Depth estimation is an important task in **computer vision**. It is used to determine the distance of an object from a camera or imaging system.

Two commonly used techniques are:

- **Stereo Vision**
- **Structure from Motion (SfM)**

The estimated and actual depth values are given below.

Object	Stereo Vision (m)	Structure from Motion (m)	Actual Depth (m)
A	2.0	2.5	2.0
B	3.0	3.8	3.0
C	4.0	5.0	4.0

PYTHON CODE	INPUT (DEPTH DATA)																																																															
<pre>1 # Q4. Depth Estimation Comparison 2 import pandas as pd 3 import numpy as np 4 5 # Input data 6 data = { 7 'Object': ['A', 'B', 'C'], 8 'Stereo_Estimated': [2.0, 3.0, 4.0], 9 'SfM_Estimated': [2.5, 3.8, 5.0], 10 'Actual_Depth': [2.0, 3.0, 4.0] 11 } 12 13 df = pd.DataFrame(data) 14 15 # Calculate absolute errors 16 df['Stereo_Absolute_Error'] = abs(df['Stereo_Estimated'] - df['Actual_Depth']) 17 df['SfM_Absolute_Error'] = abs(df['SfM_Estimated'] - df['Actual_Depth']) 18 19 # Mean Absolute Error 20 mae_stereo = df['Stereo_Absolute_Error'].mean() 21 mae_sfm = df['SfM_Absolute_Error'].mean() 22 23 # Display results 24 print("\nABSOLUTE ERROR FOR EACH OBJECT:") 25 print(df) 26 27 print("\nMEAN ABSOLUTE ERROR:") 28 print(f"Stereo Vision MAE : {mae_stereo:.3f} m") 29 print(f"Structure from Motion MAE : {mae_sfm:.3f} m") 30 31 # Comparison 32 if mae_stereo < mae_sfm: 33 print("\nRESULT: Stereo Vision is more accurate.") 34 elif mae_sfm < mae_stereo: 35 print("\nRESULT: Structure from Motion is more accurate.") 36 else: 37 print("\nRESULT: Both methods are equally accurate.")</pre>	<table><tr><th>Object</th><th>Stereo Vision Estimated Depth (m)</th><th>Structure from Motion Estimated Depth (m)</th><th>Actual Depth (m)</th></tr><tr><td>A</td><td>2.0</td><td>2.5</td><td>2.0</td></tr><tr><td>B</td><td>3.0</td><td>3.8</td><td>3.0</td></tr><tr><td>C</td><td>4.0</td><td>5.0</td><td>4.0</td></tr></table>	Object	Stereo Vision Estimated Depth (m)	Structure from Motion Estimated Depth (m)	Actual Depth (m)	A	2.0	2.5	2.0	B	3.0	3.8	3.0	C	4.0	5.0	4.0																																															
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Absolute Error Calculation

The formula for absolute error is:

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

Object A

Stereo Vision:

$$|2.0-2.0|=0 \quad |2.0-2.0|=0$$

Error = **0 m**

SfM:

$$|2.5-2.0|=0.5 \quad |2.5-2.0|=0.5$$

Error = **0.5 m**

Object B

Stereo Vision:

$$|3.0-3.0|=0 \quad |3.0-3.0|=0$$

Error = **0 m**

SfM:

$$|3.8-3.0|=0.8 \quad |3.8-3.0|=0.8$$

Error = **0.8 m**

Object C

Stereo Vision:

$$|4.0-4.0|=0 \quad |4.0-4.0|=0$$

Error = **0 m**

SfM:

$$|5.0-4.0|=1.0 \quad |5.0-4.0|=1.0$$

Error = **1.0 m**

Absolute Error Comparison

Object	Actual (m)	Stereo Estimated (m)	Stereo Error (m)	SfM Estimated (m)	SfM Error (m)
A	2.0	2.0	0.0	2.5	0.5
B	3.0	3.0	0.0	3.8	0.8
C	4.0	4.0	0.0	5.0	1.0

Mean Absolute Error

The Mean Absolute Error (MAE) is:

$$\text{MAE} = \frac{\text{Sum of Absolute Errors}}{\text{Number of Objects}}$$

Stereo Vision

$$\text{MAE} = \frac{0+0+0}{3} \text{ m} \quad \boxed{\text{MAE} = 0 \text{ m}}$$

Structure from Motion

$$\text{MAE} = \frac{0.5+0.8+1.0}{3} \text{ m} \quad \boxed{\text{MAE} \approx 0.767 \text{ m}}$$

Comparison and Evaluation

Stereo Vision produces depth values that exactly match the actual depth values.

Its errors are:

$$0 \text{ m}, 0 \text{ m}, 0 \text{ m}$$

Therefore, its MAE is:

$$\boxed{0 \text{ m}}$$

Structure from Motion produces errors of:

$$0.5 \text{ m}, 0.8 \text{ m}, 1.0 \text{ m}$$

Therefore, its MAE is approximately:

$$\boxed{0.767 \text{ m}}$$

Since a **lower error indicates better accuracy**, Stereo Vision performs better for the given data.

Result

Stereo Vision is more accurate than SfM. $\boxed{\text{Stereo Vision is more accurate than SfM.}}$

Reason: Stereo Vision has **zero absolute error**, whereas SfM has an average error of approximately **0.767 m**.

Pattern Recognition Accuracy Comparison

Introduction

Pattern recognition is an important application of **computer vision and artificial intelligence**. It involves identifying and classifying objects, images, or patterns correctly.

Two recognition systems are evaluated:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

Accuracy Calculation

The formula for accuracy is:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}} \times 100$$

Traditional System

$$\text{Accuracy} = \frac{85}{100} \times 100$$

$$\text{Accuracy} = 85\%$$

Deep Learning System

$$\text{Accuracy} = \frac{92}{100} \times 100$$

$$\text{Accuracy} = 92\%$$

PYTHON CODE

```
1 # Q5. Pattern Recognition Accuracy Comparison
2 # Data
3 traditional_correct = 85
4 traditional_total = 100
5 deeplearning_correct = 92
6 deeplearning_total = 100
7
8 # Accuracy Calculation
9 traditional_accuracy = (traditional_correct /
10                        traditional_total) * 100
11 deeplearning_accuracy = (deeplearning_correct /
12                          deeplearning_total) * 100
13
14 # Display Results
15 print("PATTERN RECOGNITION ACCURACY COMPARISON")
16 print("-----")
17 print(f"Traditional System : {traditional_correct} /
18       {traditional_total} correct")
19 print(f"Deep Learning System: {deeplearning_correct} /
20       {deeplearning_total} correct")
21 print("-----")
22 print(f"Accuracy of Traditional System :
23       {traditional_accuracy:.2f} %")
24 print(f"Accuracy of Deep Learning System :
25       {deeplearning_accuracy:.2f} %")
```

INPUT (RECOGNITION RESULTS)

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

OUTPUT (ACCURACY CALCULATION & COMPARISON)

PATTERN RECOGNITION ACCURACY COMPARISON

Traditional System : 85 / 100 correct

Deep Learning System: 92 / 100 correct

Accuracy of Traditional System : 85.00 %

Accuracy of Deep Learning System : 92.00 %

Deep Learning System is more accurate.

Accuracy Comparison

System	Correct Predictions	Total Samples	Accuracy
Traditional	85	100	85%
Deep Learning	92	100	92%

Error Rate

Traditional System

$\text{Error} = 100 - 85 = 15\%$

Deep Learning System

$\text{Error} = 100 - 92 = 8\%$

Therefore, the Deep Learning system makes fewer incorrect predictions.

Comparison and Evaluation

The Traditional system achieves an accuracy of:

85%

The Deep Learning system achieves:

92%

Difference:

$92 - 85 = 7\%$

Thus, the Deep Learning system has a **7 percentage-point higher accuracy**.

It correctly identifies 92 samples out of 100, whereas the Traditional system correctly identifies only 85 samples out of 100.

Result

Deep Learning performs better.

Reason: Deep Learning achieves **92% accuracy**, which is higher than the Traditional system's **85% accuracy**.

Precision–Recall Trade-off Analysis

Introduction

Precision and recall are important performance measures for classification and pattern-recognition systems.

The two models produce the following results:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

Where:

- **TP (True Positive):** Correctly detected positive samples
- **FP (False Positive):** Negative samples incorrectly predicted as positive
- **FN (False Negative):** Positive samples incorrectly predicted as negative

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Precision

Precision indicates how many of the samples predicted as positive were actually positive.

Formula

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Model A – Precision

Given:

$$\text{TP}=70, \text{FP}=30$$

Therefore:

$$\text{Precision} = \frac{70}{70+30} = \frac{70}{100} = 0.70 = 0.70$$
$$\text{Precision} = 70\%$$

Model B – Precision

Given:

$$\text{TP} = 85, \text{FP} = 25$$

Therefore:

$$\text{Precision} = \frac{85}{85+25} = \frac{85}{110} = 0.7727 = 0.7727$$
$$\text{Precision} \approx 77.27\%$$

Recall

Recall indicates how many of the actual positive samples were correctly detected.

Formula

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Model A – Recall

Given:

$$\text{TP} = 70, \text{FN} = 20$$

Therefore:

$$\text{Recall} = \frac{70}{70+20} = \frac{70}{90} = 0.7778 = 0.7778$$
$$\text{Recall} \approx 77.78\%$$

Model B – Recall

Given:

$$\text{TP} = 85, \text{FN} = 25$$

Therefore:

$$\text{Recall} = \frac{85}{85+25} = \frac{85}{110} = 0.7727 = 0.7727$$
$$\text{Recall} \approx 77.27\%$$

Precision and Recall Comparison

Model	TP	FP	FN	Precision	Recall
A	70	30	20	70%	77.78%
B	85	25	25	77.27%	77.27%

F1-Score

The F1-score combines precision and recall and is useful for evaluating the balance between the two.

Formula

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Model A

$$F1 = 2 \times \frac{0.70 \times 0.7778}{0.70 + 0.7778} \\ F1 \approx 73.68\%$$

Model B

$$F1 = 2 \times \frac{0.7727 \times 0.7727}{0.7727 + 0.7727} \\ F1 \approx 77.27\%$$

Comparison and Evaluation

Model A

- Precision = **70%**
- Recall = **77.78%**
- F1-score = **73.68%**

There is a difference between precision and recall:

$$77.78 - 70 = 7.78\%$$

Therefore, Model A is less balanced.

Model B

- Precision = **77.27%**
- Recall = **77.27%**
- F1-score = **77.27%**

The difference between precision and recall is:

$$77.27 - 77.27 = 0$$

Therefore, Model B has a very good balance between precision and recall.

Result

Model B is more balanced.

Reason: Model B has equal precision and recall of approximately **77.27%** and also has a higher F1-score than Model A.